Somepalli Seshagiri

Sr. Data Engineer
seshagiri.somepalli1@gmail.com || 9136017253
linkedin.com/in/somepalli-seshagiri-b50b271a5

Professional Summary:

- 11+ years of experience in ETL Architecture, Analysis, design, development, testing, implementation, maintenance and supporting of Enterprise level Data Integration, Data Warehouse (EDW), Business Intelligence (BI) solutions. Currently focused on delivering solutions using Big Data- Hadoop and its ecosystems.
- Good working experience with Echo systems like Hive, Pig, Sqoop, Map Reduce, Flume, Oozie, Kafka.
- Extensive experience in designing scalable, high-performance **data warehouse** architectures to support analytics and reporting.
- Solid Experience and understanding of implementing large scale Data Warehousing Programs and **E2E** Data integration solutions on **Snowflake, Cloud, AWS Redshift, Informatica Intelligent Cloud Services.**
- Knowledge and experience on **AWS** Services like **Redshift, Spectrum, S3, Glue, Athena, Lambda, CloudWatch,** and **EMRs like HIVE, Presto.**
- **SolrConfiguration Management** Configured and optimized Apache **Solr** for full-text search capabilities on large datasets, ensuring high availability and fault tolerance across distributed environments.
- Experience in manipulating/analysing large datasets and finding patterns and insights within structured and unstructured data.
- Proficient in developing and optimizing ETL processes using tools like Informatica, Talend, and Apache NiFi.
- Good experience in Relational Modeling and Dimensional Data Modeling using Star & Snowflake schema, De normalization, Normalization, and Aggregations.
- Experience in building pipelines using **Azure Data Factory** and moving the data into **Azure Data Lake Store.**
- Involved in developing Spark scripts using **Python** on **Azure** HDInsight for **Data** Aggregation, Validation and verified its performance over MR jobs.
- Experience in implementing **OLAP** multidimensional cube functionality using **Azure SQL Data** Warehouse.
- Managed the migration of healthcare data from legacy systems to **TriZetto platforms**, ensuring compliance with industry standards and data integrity throughout the process.
- In-depth knowledge of **Snowflake** Database, Schema and Table structures.
- Experienced configuring **Amazon EC2** instances that launch behind a load balancer, monitor the health of Amazon b instances, deploy Amazon **EC2** instances using command line calls and troubleshoot the most common problems with instances.
- Good experience in writing Pig Latin scripts, working with grunt shells and job scheduling with Oozie.
- Experience in Implementation of **CI/CD** pipeline for the Azure cloud-based analytical data ecosystem using **Azure DevOps**, **GIT** as versioning controlling and hosted pipelines for build and release
- Experience in Google Cloud components, Google container builders and **GCP** client libraries and cloud **SDK's.**
- Experienced on extending **Pig** and **Hive** core functionality by writing custom **UDF for** Data Analysis. Data transformation, file processing, and identifying user behaviour by running Pig Latin Scripts and expertise in creating Hive internal/external Tables/Views using shared Meta Store, writing scripts in HiveQL.
- Strong experience in integrating data from various sources, including relational databases, NoSQL databases, APIs, and flat files.
- Developed Hive queries help for visualizing business requirements.
- Experience in designing and implementing secure **Hadoop** clusters using **Kerberos**.

- Processing this data using Spark Streaming **API** with **Scala**.
- Implemented merge load type **ETL** solution, to load incremental data into **HDFS/Hive**.
- Excellent technical and analytical skills with clear understanding of ETL design and project architecture based on reporting requirements.
- Created efficient ETL pipelines using Databricks to ingest data from various sources and load it into data warehouses or data lakes.
- Experience in analysing data using Python, R, SQL, Microsoft Excel, Hive, PySpark, Spark SQL for Data Mining, Data Cleansing, Data Munging and Machine Learning.
- Experience in Applying Informatica latest Fixes/Patches and upgrading to new Version.
- Experience of working in all phases of SDLC. Highly skilled in Planning, Designing, developing and deploying Data Warehouses / Data Marts.
- Strong knowledge of Hadoop and Hive and Hive analytical functions.
- Implemented Proof of concepts on Hadoop stack and different big data analytic tools, migration from different databases (i.e., Teradata, Oracle, and MySQL) to Hadoop.
- Thorough understanding of Business Intelligence and Data Warehousing Concepts with emphasis on ETL.
- Proficient in designing and optimizing databases MySQL, MongoDB and SQL databases.
- Experience with Zookeeper and Kafka for monitoring and managing Hadoop jobs and using Cloudera CDH 4x, CDH 5x for monitoring and managing Hadoop clusters.
- Good experience on NoSQL technologies like HBase, Cassandra for data extraction and storing huge volumes of data. Also, experience in Data Warehouse life cycle, methodologies, and its tools for reporting and data analysis.
- Good Experience in Linux Bash scripting and following PEP-8 Guidelines in Python.
- Experience in integration of various data sources from Databases like MS Access, Oracle, SQL Server and formats like flat-files, CSV files, COBOL files and XML files.
- Experience in Performance tuning of Informatica (sources, mappings, targets and sessions) and tuning the SQL queries.
- Experience in using RDBMS concepts and worked with Oracle 10g/11g, SQL server and pleasant experience in writing stored procedures, Functions and Triggers using PL/SQL
- Proficient in **SQL, PL/SQL** and **Python** coding.
- Experience in building ETL pipelines using NIFI and good knowledge in using Apache **NiFi** to automate the data movement between different Hadoop systems.
- POC experience on **DBT** (Cleaning the data, Data Transformation (normalization, Aggregation etc.), Version Control.
- Excellent written and verbal communication skills including experience in proposal and presentation creation.

Education:

✓ Bachelor’s in computer science from Chalapathi Institute of Engineering and Technology.

Certification:

- Microsoft Certified: Azure Data Fundamentals (DP-900)
- Microsoft Certified: Azure Data Engineer (DP-203)
- Databricks Certified Data Engineer Associate

Technical Skills:

Big Data Technologies	Hadoop, Spark Core, Spark SQL, Hive, HBase, Parquet, Sqoop, Pig, Apache Kafka, Zookeeper, Flume, MapReduce Design Patterns, Data frames, Spark Streaming, Python, Hadoop Distribution (Cloudera, Horton Works), PySpark, Scala.
Data Integration & ETL	SQL, Azure Data Factory, Snowflake Modeling, Data Flow, Apache Kafka, Apache Flume, Apache NiFi, Apache Spark, Apache Beam, Apache Airflow, ELT/ETL Pipelines with Python, Snowflake Snow SQL, Spark Data Frames, Logic Apps, Azure Functions, CI/CD Frameworks (Jenkins), Data Ingestion, Azure Synapse Analytics, Polybase
Programming Languages	SQL, PL/SQL, Python, Scala.
Web Technologies	HTML, CSS, JavaScript, Restful, SOAP
Build Automation tools	Ant, Maven
Version Control	GIT, GitHub.
Databases	Azure SQL DB, Azure Synapse. MS Excel, MS Access, Oracle 11g/12c, Cosmos DB
Data Storage & Retrieval	Azure Storage, Azure SQL DB, Blob Storage, MySQL, Cassandra, Snowflake, Azure Data Lake Storage (adls), Gen2, Azure Data Bricks, Data Lakes, HDFS
Data Analysis & Reporting	Power BI, Tableau, Excel Sheets, Flat Files, CSV Files, Text files, SQL Reporting Services.
Collaboration & Project Management	Agile Methodology, CI/CD Pipelines, DevOps Collaboration.

Professional Experience:

Client: Humana, KY. **Jan 2023 to Present**

Role: Sr Data Engineer

Responsibilities:

- ✓ Engineered and implemented an automated feed file generation system for Azure Blob Storage, optimizing data organization and storage efficiency for critical information.
- ✓ Managed end-to-end ETL operations using Azure Data Factory, showcasing proficiency in orchestrating the extraction, transformation, and loading of data at large scale.
- ✓ Leveraged Azure Synapse Analytics for large-scale data processing, demonstrating competence in handling complex analytical workloads.
- ✓ Organized end-to-end ETL pipelines in Azure, seamlessly integrating Azure Databricks for large-scale big data processing, demonstrating a holistic and efficient approach to data processing and transformation.
- ✓ Pioneered real-time data processing using Azure Stream Analytics, highlighting agility in managing and analysing data streams for timely insights and decision-making.
- ✓ Spearheaded the containerization of data processing apps using Azure Kubernetes, emphasizing commitment to scalable and portable solutions.
- ✓ Implemented Apache Spark for distributed data processing, harnessing its parallel computing capabilities to efficiently handle large-scale data analytics and processing tasks.
- ✓ Leveraged Power BI for intuitive data visualization and interactive business intelligence reporting, providing actionable insights and facilitating informed decision-making within the organization.
- ✓ Developed and optimized SQL queries for efficient database operations, ensuring reliable data retrieval, manipulation, and management in support of business-critical applications.
- ✓ Implemented Slowly Changing Dimensions (SCD) techniques in data warehousing, enabling the effective tracking and management of historical changes in data, crucial for maintaining accurate and meaningful analytics over time.
- ✓ Designed and implemented a data warehouse architecture using Snowflake, leveraging its cloud-native platform to achieve scalable, elastic, and high-performance analytics with a focus on simplicity and ease of maintenance.
- ✓ Organized real-time data streaming and event-driven architectures by implementing Apache Kafka, ensuring reliable, scalable,

and fault-tolerant data pipelines for seamless communication and processing across distributed systems.

- ✓ Established and automated complex workflows by deploying Apache Airflow, enhancing data pipeline orchestration, scheduling, and monitoring for improved efficiency and reliability in data processing tasks.
- ✓ Implemented a data build tool to automate the end-to-end process of extracting, transforming, and loading (ETL) data, streamlining data workflows and ensuring consistent, reliable data outputs for analytics and reporting.
- ✓ Developed Data Definition Language (DDL) scripts for Stage and Operational Data Store (ODS) tables in Azure SQL Database, demonstrating expertise in database design and meticulous management for efficient data storage and retrieval.
- ✓ Led the development of DBT models, macros, and data marts in Snowflake or Azure Synapse Analytics, showcasing proficiency in crafting efficient and structured data models to support advanced analytics and reporting needs.
- ✓ Innovated by crafting custom activities using Azure Functions and PowerShell scripts, streamlining processes, and significantly enhancing operational efficiency through automation and tailored solutions.
- ✓ Implemented automation using Azure Logic Apps, with integration into Azure Key Vault for enhanced security measures, data governance.
- ✓ Implemented Apache Airflow in Azure for HiveQL scripts and jobs, orchestrating complex workflows for seamless execution.
- ✓ Spearheaded the implementation of CI/CD pipelines and frameworks with Azure DevOps, ensuring a streamlined and automated development lifecycle.
- ✓ Actively participated in Agile ceremonies and collaborated with DevOps engineers, utilizing JIRA and Azure DevOps for effective project management and collaboration.
- ✓ Documented technical specifications, data flow diagrams, and process documentation for Azure Architecture, ensuring clear and comprehensive documentation for knowledge transfer and future reference.

Environment: Azure SQL Database, Azure Data Factory, Azure Synapse Analytics, Azure Databricks, Azure Kubernetes, Azure DevOps, ADLS Gen2, Event Hubs, PolyBase, PySpark, HDFS, MapReduce, Spark, Spark Streaming, Kafka, Hive, ORC, Parquet, Text File, Compression Techniques, Dimensional Modeling, Star Schema, Snowflake Schema, SQL Server, Oracle, Teradata.

Client: Albertsons, TX. **Mar 2020 to Dec 2022**
Role: Azure Data Engineer
Responsibilities:

- Experience working with cross-functional teams to gather requirements, design and implement data solutions, and provide technical support.
- Experience working with Azure data services such as Azure Data Factory, Azure Databricks, Azure Data Lake, E-commerce Azure Synapse Analytics and Azure SQL.
- Experience in designing, developing, and maintaining data pipelines using Azure Data Factory, streaming data from various sources like IoT devices, logs, social media, etc.
- Expertise in using Azure Databricks or any other data transformation technologies to transform raw data into meaningful insights.
- Experience in designing, building, and deploying data models.
- Expertise in working with different types of data sources, such as Azure Blob Storage, Azure Data Lake Storage, E-commerce, Azure SQL Database, and Azure Synapse Analytics, as well as on-premises data sources such as SQL Server and Oracle.
- Experience in building complex data integration workflows using Azure Data Factory, including data mapping-commerce data transformation, and data cleansing.
- Understanding of Data Factory integration with other Azure services, such as Azure Functions, Azure Databricks, and Azure Stream Analytics.
- Proficiency in data ingestion, integration, and transformation using Azure Databricks with various data sources, including Azure Blob Storage, E-commerce, Azure Data Lake Storage, Azure SQL Database, and Azure Synapse Analytics.
- Developed Tabular cubes in SSAS Connecting to Azure Synapse Analytics.
- Developing visual reports, dashboards and KPI scorecards using Power BI desktop.
- Involved in writing DAX expressions for Power BI Calculated fields.
- Responsible for Power BI Modeling from various data sources and enabling the datasets to end users.

- Designed and optimized Snowflake schemas, tables, and views to support complex business intelligence queries and data transformations.
- Created and maintained ETL pipelines using Snowflake built-in features and tools, automating data ingestion, transformation, and loading processes.
- Collaborated with cross-functional teams to integrate Snowflake with various data sources, such as Salesforce, and internal systems, ensuring data accuracy and consistency.
- Implemented Snowflake security measures, including role-based access control (RBAC), to protect sensitive data and ensure compliance with privacy regulations.
- Experience in mentoring junior data engineers to equip them with the capabilities to function independently on client assignments.

Environment: Azure Databricks, Azure Data Factory, Azure Logic Apps, Functional Apps, MySQL, Oracle, HDFS, MapReduce, Apache Spark, Apache Hive, Python, Scala, PySpark, PostgreSQL, .NET, Teradata, Shell Scripting, GIT, JIRA, Jenkins, Apache Kafka, Power BI, Azure Synapse Analytics, Polybase, Azure Data Lake Storage, Azure SQL Database, Azure DevOps.

Client: Pitney Bowes Inc., CT **Jan 2017 to Mar 2020**
Role: Data Engineer
Responsibilities:

✓ Design & implement migration strategies with Azure suite: Azure SQL Database, Azure Data Factory (ADF) V2, Azure key Vault, Azure Blob Storage.
✓ Created end-to-end Data pipelines using ADF services to load data from On-Prem to Azure SQL server for Data orchestration.
✓ Build scalable and reliable ETL pipelines to pull large and complex data from different systems efficiently.
✓ Worked on building the data pipeline using Azure Service like Data Factory to load the data from Legacy, SQL server to Azure Data warehouse using Data Factories and Databricks Notebooks.
✓ Develop Spark applications using PySpark, spark SQL for data extraction, transformation, and aggregation from multiple file formats for analysing, and transform.
✓ Experience with RESTful API development, including defining resources, handling requests, and returning responses in JSON or XML formats.
✓ Expertise you have with building Kafka-based data pipelines, including data ingestion, processing, and analysis.
✓ Experience with Kinesis APIs, including Kinesis Client Library and Kinesis Producer Library.
✓ Proficiency with Python libraries commonly used in data engineering, such as Pandas, NumPy, PySpark, and Scikit-learn.
✓ Expertise with using Pub/Sub messaging systems in combination with big data technologies, such as Hadoop, Spark, or Flink.
✓ Experience you have with building CDC pipelines, including designing schema changes, capturing changes in real-time, and applying changes to downstream systems.
✓ Developed Python based API (RESTful Web Service) using Flask. Involved in Analysis, Design, and Development and Production phases of the application.
✓ Developed Python batch processors to consume and produce various feeds.
✓ Constructed product-usage data aggregations using Py-Spark, Spark SQL and maintained in Azure Data warehouse for reporting, data science dashboarding and ad-hoc analyses.
✓ Leveraged Jenkins for continuous integration services.
✓ Involved in AJAX driven applications by invoking web services/API and parsing the JSON response.
✓ Created a Git repository and added the project to GitHub.
✓ Utilized Agile process and JIRA issue management to track sprint cycles.

Environment: Azure SQL Database, Azure Data Factory (ADF) V2, Azure Key Vault, Azure Blob Storage, PySpark, Spark SQL, RESTful API, Kafka, Kinesis APIs, Python libraries (Pandas, NumPy, Scikit-learn), Pub/Sub messaging systems, Hadoop, Spark, Flink, CDC pipelines, Flask, Jenkins, AJAX, JSON, Git, GitHub, Agile, JIRA.

Client: COX, Atlanta **Nov 2013 to Dec 2016**
Role: Data Engineer
Responsibilities

- Troubleshooting Spark applications for improved error tolerance.
- Leveraged AWS Glue and AWS Lambda alongside Apache Spark for advanced data transformations, complemented by meticulous data profiling and preparation facilitated by AWS Glue DataBrew to maintain exceptional data quality.
- Transformed and serialized data into Apache Avro and Parquet formats using AWS Lambda, and efficiently stored it in Amazon S3, ensuring optimal storage, scalability, and data accessibility.
- Implemented data quality checks and ongoing validation using AWS Glue's built-in data quality features, ensuring the accuracy and consistency of data throughout the ETL pipeline.
- Managed data processing workflows efficiently using AWS Step Functions to orchestrate and automate ETL pipelines, ensuring seamless data processing and transformation.
- Utilized AWS Glue Data Brew to automate data preparation and seamlessly loaded transformed data into Amazon Redshift, optimizing structured data storage for efficient analytics.
- Implemented real-time monitoring and issue identification during data migration using Amazon CloudWatch, while ensuring comprehensive auditing and logging through AWS CloudTrail for enhanced data management and security.
- Enhanced ETL job performance by fine-tuning AWS Glue job parameters, optimizing parallelization, and leveraging AWS EMR (Elastic MapReduce) for large-scale data processing and transformations, resulting in significant performance improvements.
- Implemented AWS CloudWatch to conduct comprehensive performance testing and monitor the health and efficiency of AWS resources
- Collaborated with AWS Solutions Architects to design and implement AWS Auto Scaling strategies for seamless resource scalability, optimizing application performance and cost-efficiency.
- Implemented role-based access control using AWS Identity and Access Management (IAM) to ensure secure and granular access to AWS resources and employed AWS Key Management Service (KMS) for end-to-end encryption, safeguarding data at rest and in transit.
- Managed and safeguarded sensitive credentials and encryption keys effectively by utilizing AWS Secrets Manager, ensuring robust data security and compliance within AWS environments.
- Implemented a robust CI/CD pipeline using AWS Code Pipeline to automate and streamline the deployment of ETL jobs and data pipelines, ensuring rapid and consistent software delivery.
- Implemented Agile practices using AWS DevOps, fostering cross-functional collaboration and iterative development, resulting in enhanced efficiency and rapid project execution within AWS-based data engineering projects.

Environment: PySpark, Python, Hive, Tableau, AWS Lambda, EMR, Glue, step function, Cloud watch, Shell- scripting, Linux, Jenkins, Bitbucket.

Client: Incresol, India. **Aug 2012 to Sep 2013**

Role: Big Data Developer

Responsibilities

- ✓ Design and implement efficient data ingestion pipelines to collect structured and unstructured data from various sources, such as customer databases, third-party data providers, social media, and IoT devices.
- ✓ Develop ETL processes to clean, transform, and load the incoming data into the company's data lake or data warehouse.
- ✓ Setting up and maintaining scalable and distributed data storage solutions, such as Hadoop Distributed File System (HDFS).
- ✓ Implement data partitioning and compression strategies to optimize storage and retrieval performance.
- ✓ Ensure data security and compliance with industry regulations, such as HIPAA and GDPR, through access controls, data encryption, and data governance policies.
- ✓ Develop and optimize big data processing pipelines using frameworks like Apache Spark, Apache Flink, or Apache Kafka.
- ✓ Performing data cleaning, transformation, and analysis tasks on large datasets to extract valuable insights.
- ✓ Develop predictive models for risk assessment, customer churn analysis, fraud detection, and claims processing automation.
- ✓ Create interactive dashboards and reports using tools like Power BI.
- ✓ Presenting data-driven insights and recommendations to various stakeholders, including executives, underwriters, and claims analysts.
- ✓ Define and implement data governance policies and procedures to ensure data quality, consistency, and lineage.
- ✓ Maintain and update metadata repositories to document data sources, transformations, and usage.
- ✓ Collaborate with data stewards and data governance teams to ensure compliance with internal and external regulations.
- ✓ Optimize data pipelines, queries, and algorithms for better performance and scalability.
- ✓ Implement caching, indexing, and partitioning strategies to improve query response times.
- ✓ Provisioning and managing cloud resources (e.g., EC2 instances, EMR clusters) for big data workloads.
- ✓ Implement Infrastructure as Code practices and automated deployment pipelines for efficient and repeatable infrastructure management.
- ✓ Staying up to date with the latest big data technologies, tools, and best practices through research, training, and professional development.
- ✓ Participating in industry events, conferences, and communities to learn from peers and share knowledge.
- ✓ Working closely with data engineers, data analysts, business stakeholders, and cross-functional teams to understand their requirements, provide technical guidance, and ensure successful project delivery.
- ✓ Maintain comprehensive documentation for data pipelines, models, and processes to ensure maintainability and knowledge transfer.
- ✓ Contribute to internal knowledge bases, wikis, or code repositories to share best practices and reusable components.

Environment: Apache Spark, Scala, PySpark, Sqoop, Hive, MapReduce, Oracle, MySQL, Cassandra, HBase, HDFS, Spark Streaming, Kafka, PowerBI, YAML, Git, GitHub, Zookeeper.